DESIGN PROJECT · IIIT VADODARA

# LLM Hallucination Detection using Multi-Layer EigenScore

Catching an AI that is confidently wrong, using only what happens inside the model.

PROBLEM STATEMENT

Large language models often produce **fluent, confident answers that are factually wrong** (hallucinations), in the same tone as correct ones.

We need a detector that flags these answers **without knowing the right answer**, without external fact-checking, and that runs on an **ordinary 8 GB laptop GPU**.

**"In which year did the bridge open?" asked 5 times:**

1932 1932 1932 1932 1932

knows the answer: responses agree

1932 1928 1936 1941 1930

guessing: responses scatter (illustrative)


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## 01 How we solved it

1 **Ask again**

Sample **10 answers** to the same question at moderate randomness.

2 **Look inside**

For each answer, read the model's internal state: middle layers, last token, **4096 numbers**.

3 **Measure spread**

Covariance of the 10 answers → **eigenvalues** → one EigenScore. High = likely wrong.

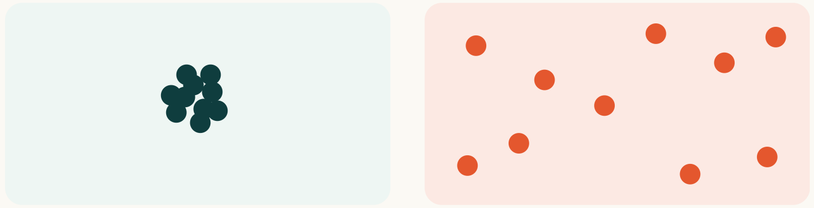
4 **Calm overconfidence**

**Feature clipping**: cap rare extreme activations so confident errors start to vary.

5 **Fit a laptop**

**4-bit** model, one-layer capture, 8 automated tests: runs on an **8 GB** GPU.

### Spread in meaning = uncertainty



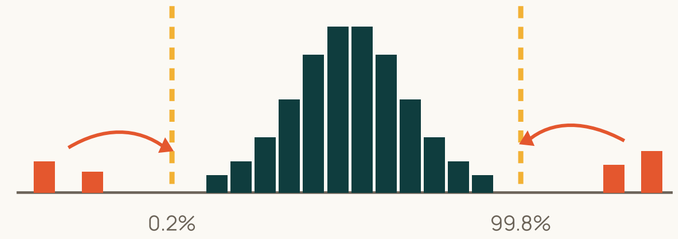
tight → low score → trust

spread → high score → flag

$$\text{EigenScore} = \text{mean}\left( \log \lambda_i \right), \quad \lambda = \text{eig}\left( \text{Cov}(\text{answers}) + \alpha I \right)$$

**Multi-Layer EigenScore** averages this score over a band of middle layers (8–24 of 32) instead of one hand-picked layer, for a steadier, model-independent signal.

### Feature clipping during generation




Each neuron of the penultimate layer is capped to its 0.2–99.8 percentile range, estimated from the last **3,000 tokens**. Extreme "shouting" values (coral) are pulled in, reducing over-confident generation.

## 02 Results

**83.5 AUROC**

Hallucination detection accuracy of our system

Chance that a wrong answer is ranked more suspicious than a correct one (50 = coin toss, 100 = perfect). With feature clipping enabled: 82.5.




**250** test questions (CoQA reading comprehension)

**10** sampled answers per question

**8 GB** laptop GPU (RTX 4060), 7B model in 4-bit

**~6 s** per question, end to end

**0.29%** activations clipped, as designed

**8 / 8** automated correctness tests passing

ONGOING Evaluating **Multi-Layer EigenScore**, scale normalisation and a confidence-aware hybrid on 1,000 questions each across 3 QA datasets.


EigenScore metric: Chen et al., "INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection," ICLR 2024.


Model: Llama-2-7B · Dataset: CoQA